Lyapunov Resource Control for Correlated Multi-Agent Markov Learning

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***Abstract*—Identifying dependence consumes the resources that could otherwise train a learner. We study when this expense improves terminal risk beyond fixed sampling under message and environment-clock budgets. A discrete Lyapunov score couples participation, spacing, contraction, correlated noise, staleness and affordable horizon for affine temporal-difference learning. For a Gaussian Markov subclass, same-stream risk bounds account for probe decisions and their dependence on the continuing learner. Target-invariant contrasts make the probe observable without knowing the signal level. Within a finite linear-program class, an initialization-minimax characterization separates covariance-regime information from current-state information. For stable averaged learners, extra state value is second order in budget, and a two-program construction proves this order sharp. Covariance identification can instead improve the leading risk coefficient against every affordable prefix of each constant resource program. A construction for running means and averaged scalar temporal-difference critics remains uniformly advantageous under variance, persistence and prior uncertainty. At a specified budget, their risk guarantees improve on every predetermined resource schedule by at least 2.54 and 1.86 percent, respectively, with all costs paid. Simulations quantify the protection obtained by declining uneconomic probes and the gains from resource-matched learning. These results provide constructive conditions for paid identification to improve risk and delimit the comparisons for which that advantage is established.**

***Index Terms*—Multi-agent learning, Markov data, temporal-difference learning, adaptive sensing, Lyapunov methods.**

I. Introduction

Policy evaluation estimates the discounted return of a fixed policy, supplying the critics and advantages used by actor–critic systems. We study parallel actors executing a common target policy on Markov environment copies and sending temporal-difference observations to a value-learning server. The server chooses participation q, physical sample spacing b, and step size η under finite message and environment-clock budgets.

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Parallel samples need not provide parallel information: actors can share simulator randomness or exogenous disturbances while preserving their marginal transition laws. At correlation $\rho > 0$, equicorrelated Gaussian observations provide effective participation $q/[1 + (q - 1)\rho] \leq 1/\rho$, while messages grow with q. Larger spacing suppresses Markov bias but consumes transitions; larger steps accelerate contraction but raise noise and stale-update risk. The design must price these effects against the affordable learning horizon and dependence-identification cost.

Temporal-difference and stochastic-approximation theory [1]–[4] provides finite-time analyses under Markov sampling [5]–[7], concentration bounds [8], and Lyapunov-based adaptive learning rates [9]. Participation and spacing also determine the observation stream.

Distributed/federated learning studies consensus, parallelism, coverage, and sample–communication tradeoffs [10]–[15]. Clustered clients, adaptive batches, and antithetic sampling regulate variance [16]–[18]; feedback controls participation [19], while convergence-bound-driven Lyapunov control selects clients and powers [20]. We ask how covariance and spacing change error per update and the number of affordable updates.

Asynchronous optimization analyzes stale computation [21]–[23]; delay-adaptive stochastic approximation [24], [25] and stochastic delay differential equations (SDDEs) support convergence and worker-activation design [26]. Here delay, cross-agent covariance, and paid spacing jointly determine terminal risk. DASA [25] permits unbounded delays and obtains an average-delay rate, but its cross-agent independence assumption removes the common-source covariance considered here. Lyapunov client/power scheduling [20] minimizes a convergence/time surrogate under average power constraints; SDDE worker activation [26] optimizes a delay-dependent convergence approximation. Our distinction is a paid identification decision evaluated through the remaining learner's terminal risk on the continuing stream.

Sequential design and identification supply change-of-measure/allocation tools [27]–[33], with conservative bandits constraining baseline-relative reward [34]. We value identification through its effect on the remaining learner's risk. Covariance-adaptive identifica-

tion [32] minimizes joint-query complexity for independent rounds, rather than the downstream learner's risk after paying those queries. Parameter-free TD [35] addresses unknown step-size constants, and adaptive Markov SA [36] controls all-future trajectory excursions under kernel regularity. Classical Bayesian design [37] and adaptive-data analysis [38] already price information through decision utility and selection effects. Joint detection and estimation couples posterior estimation loss to decisions [39], [40]; sequential procedures also price observation cost and control estimation and detection errors [41], including robust formulations [42]. We use classical error contrasts [43] to make calibration observable without the unknown target and classical minimax interchange [44] for the vector initialization extension. The resource-program analysis distinguishes a second-order value of the current noise state from a possible first-order benefit of covariance identification, compared with every affordable fixed training prefix. Our Gaussian refinement uses classical association [45] and truncated-normal moments [46]; its specific object is the signed risk of a paid probe followed by the same continuing learner, rather than a new general selection or information-value principle.

The main contributions are as follows.

- **Finite-Budget Lyapunov Resource Design:** A terminal-risk score couples participation, physical Markov spacing, and step size to the number of affordable updates under dual budgets.
- **Learning and Certificate Guarantees:** An affine delayed Markov bound supplies the resource score and critic-tracking guarantee under uniform-history conditions. Predictable moment and mixing bounds control calibration-induced selection error.
- **When Paid Identification Is Useful:** Same-stream risk transfer prices identification through the continuing learner. The current-state value has a sharp second-order ceiling, whereas regime information can improve the leading coefficient. A resource-envelope bound and an explicit TD filter correction establish finite-budget separation from every predetermined schedule over covariance and prior uncertainty.
- **Resource-Matched Learning Evidence:** Controlled studies quantify risk and area under the curve (AUC) gains from finite-horizon control. Held-out multi-agent proximal policy optimization (MAPPO) studies test participation control, and observable-probe simulations evaluate economic admission against indiscriminate probing and constant-program prefixes.

The remainder of the paper develops the argument as follows. The formulation and resource geometry lead to the finite-time Lyapunov controller, followed by paid-identification guarantees and the state-versus-regime value characterization. Implementation interfaces and simulations then connect these guarantees to the evaluated learning and resource comparisons.

II. Model and Finite-Budget Objective

A. Fixed-policy evaluation with multiple actors

Let $(\mathcal{S}, \mathcal{U}, P, r, \gamma)$ be a discounted Markov decision process and let $\pi_{\mathrm{tar}}(u \mid s)$ be a fixed target policy. Under π_{tar}, the state–action process is a Markov reward process with transition kernel P_π and stationary distribution π. For a feature vector $\phi(s) \in \mathbb{R}^d$, the linear value approximation is $V_w(s) = \phi(s)^\top w$. Its projected Bellman fixed point $w^\star$ solves

$$Aw^\star = c, \quad A = \mathbb{E}_\pi[\phi_t(\phi_t - \gamma\phi_{t+1})^\top], \quad c = \mathbb{E}_\pi[r_t\phi_t]. \tag{1}$$

Let S_k be the start of retained transition k and $X_{i,k} = (s_{i,S_k}, r_{i,S_k}, s_{i,S_k+1})$. The server chooses $q_k \in \{1, \dots, Q\}$ actors and start-to-start spacing $b_k \in \mathbb{N}$. For temporal-difference data, P_q is a Markov augmentation containing the joint transition tuples, not the raw state kernel; retained starts are separated by b_k kernel steps. Its invariant law π_q satisfies the total-variation (TV) bound

$$\sup_z \left\| P_q^b(z, \cdot) - \pi_q \right\|_{\mathrm{TV}} \le C_q r_q^b, \quad 0 < r_q < 1. \tag{2}$$

Every actor has the target-policy marginal law; cross-agent dependence enters the aggregate stationary moments.

For the identification layer, a Gaussian common-factor subclass makes the dependence parameter explicit. Under instance $j \in \{0, 1\}$,

$$\begin{aligned} C_{S_k} &= \lambda^{b_k} C_{S_{k-1}} + \sqrt{1 - \lambda^{2b_k}}\, \xi_{j,k}, \\ X_{i,k} &= C_{S_k} + \epsilon_{i,k}, \end{aligned} \tag{3}$$

where $\xi_{j,k} \sim \mathcal{N}(0, \theta_j)$, $\epsilon_{i,k} \sim \mathcal{N}(0, 1)$, and all private innovations are independent. The common process is initialized with $C_{S_0} \sim \mathcal{N}(0, \theta_j)$, so fixed-clock probes are stationary. Under adaptive sampling, the conditional predictive law determines the information in each observation. The corresponding same-time correlation is $\rho_j = \theta_j/(1 + \theta_j)$. Within each instance, every actor has the same marginal law; changing θ_j changes both marginal variance and cross-agent correlation. This Gaussian subclass is used for the information boundary; the convergence theorem applies to bounded affine temporal-difference observations satisfying Assumption 1.

B. Affine delayed temporal-difference recursion

Let $w^\star \in \mathbb{R}^d$ be the projected temporal-difference fixed point and $e_k = w_k - w^\star$. For action $a = (q, b, \eta)$ and delays $0 \le \tau_i \le D$, the retained recursion is

$$e_{k+1} = e_k - \frac{\eta}{q} \sum_{i=1}^{q} \left(H_{i,k} e_{k-\tau_i} - \xi_{i,k} \right). \tag{4}$$

The rank-one matrix $H_{i,k}$ is the sample temporal-difference Jacobian and $\xi_{i,k}$ is the affine innovation at $w^\star$. For temporal-difference (TD) learning with zero trace, TD(0), these quantities are

$$H_{i,k} = \phi_{i,k}(\phi_{i,k} - \gamma\phi'_{i,k})^\top, \quad \xi_{i,k} = \phi_{i,k}\delta_i(w^\star), \quad (5)$$

where $\delta_i(w) = r_{i,k} + \gamma\phi'_{i,k}{}^\top w - \phi_{i,k}^\top w$. The first tuple must satisfy the same conditional mixing bound, or be initialized independently from stationarity.

Assumption 1 (Certified action domain)**.** *The public sets $\mathcal{Q} \subseteq \{1, \dots, Q\}$ and $\mathcal{B} \subset \mathbb{N}$ are finite, while the step interval $[\eta_{\min}, \eta_{\max}]$ is compact. For each $a = (q, b, \eta)$ in their product: (a) $\|H_{i,k}\| \le L$ and $\|\xi_{i,k}\| \le G$ almost surely; (b) $\mathbb{E}_{\pi_q}\xi_{i,k} = 0$, and the common mean $A = \mathbb{E}_{\pi_q}H_{i,k}$ has symmetric part at least μI; (c) the delay profile is deterministic, satisfies $0 \le \tau_i \le D$, and is known when scoring; and (d) predictable upper bounds cover the stationary aggregate moments and the mixing law uniformly conditional on pre-observation history. Initialize $w_s = w_0$ for all $s \le 0$; increments before the first update are zero.*

Write $\mathcal{A} = \mathcal{Q} \times \mathcal{B} \times [\eta_{\min}, \eta_{\max}]$; integer actor counts and clocks are enumerated, with a continuous scalar step search.

C. Dual budgets and terminal risk

Each scheduled block pays $h + q$ message units and b parallel-clock units, including unused padding within the block. Let $u \ge 0$ count terminal scheduled updates excluded from the reported learner and let $\ell \ge 0$ be a separately declared physical startup/drain charge. The recursion's state-lag bound D is separate from these terminal charges; any equality is part of the declared protocol. A feasible trajectory therefore satisfies

$$\sum_k (h + q_k) \le B_{\mathrm{msg}}, \quad \sum_k b_k + \ell\mathbf{1}\{K > 0\} \le B_{\mathrm{env}}. \quad (6)$$

For a stationary action, all scheduled blocks are paid, and the usable horizon is

$$N_a(B) = \left[\min\left\{\left\lfloor \frac{B_{\mathrm{msg}}}{h+q} \right\rfloor, \left\lfloor \frac{B_{\mathrm{env}} - \ell}{b} \right\rfloor\right\} - u\right]_+. \quad (7)$$

The controller minimizes a finite-time certificate $\overline{R}_j(a; B, D)$ satisfying $\mathbb{E}_j\|e_{N_a}\|^2 \le \overline{R}_j(a; B, D)$. This parameter mean-square risk controls the value-estimation risk of $J_\nu(w) = \mathbb{E}_{s\sim\nu}V_w(s)$, as quantified by Corollary 1.

The Gaussian running-mean learner used in the mechanism study starts from a unit pseudo-observation. For $r = \lambda^b$, its exact risk is

$$R_j^{\mathrm{mean}}(a, N) = \frac{1 + N/q}{(N+1)^2} + \frac{\theta_j\left[N + 2\sum_{s=1}^{N-1}(N-s)r^s\right]}{(N+1)^2}. \quad (8)$$

Eq. (8) exposes opposing effects: larger q reduces private noise but can reduce N, while spacing controls r at a transition cost.

III. Correlation and Resource Geometry

A. A sharp spatial speedup ceiling

The statistical role of participation is exact in a Gaussian location subclass. Let $Y_{i,t} = \vartheta + C_t + \epsilon_{i,t}$ with common variance σ_c^2, private variance σ_e^2, and independent rounds. For a fixed participation q, the minimax squared-error risk after T rounds is

$$R_q(T) = \frac{1}{T}\left(\sigma_c^2 + \frac{\sigma_e^2}{q}\right). \quad (9)$$

Writing $\rho = \sigma_c^2/(\sigma_c^2 + \sigma_e^2)$ gives the speedup relative to one actor,

$$S(q, \rho) = \frac{R_1(T)}{R_q(T)} = \frac{q}{1 + (q-1)\rho} \le \min\{q, \rho^{-1}\}. \quad (10)$$

Thus information saturates at positive correlation while messages grow with q.

Proposition 1 (Participation phase under a message budget)**.** *Suppose the environment budget does not bind, $h > 0$, $0 < \rho < 1$, and a message budget B_{msg} is used with fixed q. Ignoring integer rounding, the minimax risk is proportional to*

$$F(q; \rho, h) = (h + q)\left(\rho + \frac{1-\rho}{q}\right). \quad (11)$$

Its continuous minimizer is

$$q^\star_{\mathrm{cont}} = \sqrt{\frac{h(1-\rho)}{\rho}}, \quad (12)$$

clipped to $[1, Q]$, and an integer optimum is one of its two neighboring feasible participation levels.

Differentiating F gives $\partial_q F = \rho - h(1-\rho)/q^2$, proving Proposition 1. Fixed overhead favors larger batches; common randomness favors smaller participation.

B. Temporal spacing and the usable horizon

For the common-factor process in Eq. (3), retaining every bth observation changes persistence from λ to $r = \lambda^b$. The integrated correlation term in Eq. (8) admits the closed form

$$\sum_{s=1}^{N-1}(N-s)r^s = \frac{r\{(N-1)-Nr+r^N\}}{(1-r)^2}. \tag{13}$$

For nonnegative persistence, increasing b suppresses the positive correlation term, but Eq. (7) decreases approximately as $1/b$ when the environment clock binds. Consequently, aggressive thinning can make retained observations nearly independent and still increase terminal error.

The next section jointly prices noise and contraction over this affordable horizon.

IV. Lyapunov-Certified Joint Resource Control

A. Action-dependent one-block drift

The base Lyapunov function is the squared parameter error $V(e) = \|e\|^2$. Delay prevents its one-step drift from closing in e_k alone because the update also contains $e_{k-\tau_i}$. We therefore analyze the observable-independent sliding envelope

$$R_k = \max_{k-2D\le s\le k}\mathbb{E}V(e_s), \tag{14}$$

which is a Lyapunov–Razumikhin construction for the lifted history $(e_k, e_{k-1}, \dots, e_{k-2D})$.

Define the aggregate stationary moments

$$K_q = \lambda_{\max}\big(\mathbb{E}_{\pi_q}[\bar{H}_k^\top \bar{H}_k]\big), \quad \Omega_q = \mathbb{E}_{\pi_q}\|\bar{\xi}_k\|^2, \tag{15}$$

where $\bar{H}_k = q^{-1}\sum_i H_{i,k}$ and $\bar{\xi}_k = q^{-1}\sum_i \xi_{i,k}$. Let $\delta = C_q r_q^b$ and $\mu_\delta = \mu - 2L\delta$. For the delay root-mean-square $\tau_{\rm rms} = (q^{-1}\sum_i \tau_i^2)^{1/2}$, define

$$\begin{aligned} a_\delta(\eta) &= 1-\eta\mu_\delta+\eta^2(2K_q+4L^2\delta),\\ \beta_\delta(\eta) &= \frac{4\eta G^2\delta^2}{\mu_\delta}+\eta^2(2\Omega_q+4G^2\delta),\\ h_\tau(\eta) &= 2\eta^4L^4\tau_{\rm rms}^2, \quad g_\tau(\eta) = 2\eta^4L^2G^2\tau_{\rm rms}^2. \end{aligned} \tag{16}$$

For positive delay, set $\lambda_\star = \sqrt{h_\tau/a_\delta}$ and

$$\begin{aligned} c_{\rm aff} &= (1+\lambda_\star)a_\delta + (1+\lambda_\star^{-1})h_\tau,\\ d_{\rm aff} &= (1+\lambda_\star)\beta_\delta + (1+\lambda_\star^{-1})g_\tau. \end{aligned} \tag{17}$$

The zero-delay definitions are the continuous limits $c_{\rm aff} = a_\delta$ and $d_{\rm aff} = \beta_\delta$. These coefficients separate mean contraction from sampling noise, finite-mixing bias, and the fourth-order delay terms.

Theorem 1 (Affine delayed Markov Lyapunov bound)**.** *Suppose Assumption 1 holds, $\mu_\delta > 0$, $a_\delta(\eta) > 0$, and $c_{\rm aff}(a) < 1$. Let $m = 2D+1$ and $R_k = \max_{k-2D\le s\le k}\mathbb{E}\|e_s\|^2$. Then, for every integer $n \ge 0$,*

$$R_{nm} \le c_{\rm aff}(a)^n R_0 + \frac{d_{\rm aff}(a)}{1-c_{\rm aff}(a)}. \tag{18}$$

Supplementary Appendix A proves the bound: correlation enters (K_q, Ω_q), persistence enters δ, and delay changes contraction and forcing.

Corollary 1 (Finite-horizon return certificate)**.** *Under Theorem 1, let $n = \lfloor N_a(B)/(2D+1)\rfloor$ and let $\psi_\nu = \mathbb{E}_{s\sim\nu}\phi(s)$. Then*

$$\mathbb{E}\,|J_\nu(w_{N_a}) - J_\nu(w^\star)|^2 \le \|\psi_\nu\|^2\left[c_{\rm aff}^n R_0 + \frac{d_{\rm aff}}{1-c_{\rm aff}}\right]. \tag{19}$$

Supplementary Appendix A-B handles incomplete terminal blocks and connects parameter risk to fixed-policy value-estimation error.

The same certificate also tracks a critic target moved by a policy learner. Let $w_r^\star$ be the projected fixed point during epoch r, lasting one replacement block $m = 2D+1$, and let $Z_r = \max_{rm-2D\le s\le rm}\mathbb{E}\|w_s - w_r^\star\|^2$.

Theorem 2 (Moving-policy critic tracking)**.** *Suppose every epoch satisfies Assumption 1, possibly with a different certified action a_r, and its coefficients obey $c_{\rm aff}(a_r) \le c < 1$ and $d_{\rm aff}(a_r) \le d$. Require the fixed-target one-step inequality $\mathbb{E}\|w_{k+1} - w_r^\star\|^2 \le c\max_{k-2D\le s\le k}\mathbb{E}\|w_s - w_r^\star\|^2 + d$ throughout epoch r, including its mixed-history boundary. This requirement holds for a consistently initialized epoch with no carried old-target updates; without that protocol it is an additional uniform-history condition. If the policy-induced target moves only between epochs and $\|w_{r+1}^\star - w_r^\star\| \le v$, then for every $\chi > 0$ satisfying $\bar{c} = (1+\chi)c < 1$,*

$$Z_n \le \bar{c}^n Z_0 + \frac{(1+\chi)d + (1+\chi^{-1})v^2}{1-\bar{c}}, \tag{20}$$

$$\mathbb{E}|\widehat{A}_w - A^\star|^2 \le (1+\gamma)^2\Phi^2 Z_n, \quad \|\phi(s)\| \le \Phi. \tag{21}$$

The bound exposes a moving-target phase: policy improvement increases v, while the resource action determines (c, d). Supplementary Appendix A-C proves the positive-part recursion. Under the uniform-history condition, recomputing the joint score at actor-epoch boundaries controls the certified critic lag.

B. Joint action selection

For $n_a = \lfloor N_a(B)/m\rfloor$, the certificate score is

$$\mathcal{U}_j(a; B, D) = R_\star(a) + c_{\rm aff}(a)^{n_a}\left(R_0 - R_\star(a)\right)_+, \tag{22}$$

where $R_\star(a) = d_{\mathrm{aff}}(a)/(1-c_{\mathrm{aff}}(a))$. The certified oracle for a known dependence instance is

$$a_j^\star(B, D) \in \arg\min_{a\in\mathcal{A}} \mathcal{U}_j(a; B, D). \tag{23}$$

For every pair (q, b), define the certified scalar domain

$$\mathcal{E}_{q,b} = \{\eta \in [\eta_{\min}, \eta_{\max}] : \mu_\delta \geq \mu_0 > 0, \\ a_\delta(\eta) \geq a_0 > 0,\ c_{\mathrm{aff}}(q, b, \eta) \leq 1 - \kappa\}, \tag{24}$$

The controller solves

$$\eta^\star_{q,b} \in \arg\min_{\eta\in\mathcal{E}_{q,b}} \mathcal{U}_j(q, b, \eta; B, D), \\ (q^\star, b^\star) \in \arg\min_{q,b} \mathcal{U}_j(q, b, \eta^\star_{q,b}; B, D). \tag{25}$$

Here $\eta_{\min} > 0$, and μ_0, a_0, κ are declared positive certification margins. Pairs with empty scalar domains are omitted; at least one nonempty domain is required, otherwise the controller returns a declared feasible fallback or performs no update. The optimization defines global scalar minima; numerical candidates attain this oracle guarantee when their global accuracy is certified.

Proposition 2 (Existence and computation)**.** *If K_q, Ω_q, and the mixing certificates are finite and the declared certified domain is nonempty, then $\mathcal{U}_j$ is continuous on its compact scalar domains and attains a minimum. With $N_q = |\mathcal{Q}|$, $N_b = |\mathcal{B}|$, and L_η scalar evaluations per search, the certificate cost is $O(N_q N_b L_\eta)$ arithmetic after the moment table is cached. For rank-one temporal-difference samples, update aggregation costs $O(qd)$ arithmetic and $O(d)$ working memory per retained block; exact spectral moment calibration is a separate cost.*

Supplementary Appendix A-D proves compactness and separates aggregation from spectral calibration.

C. Finite-horizon and one-step control

A purely myopic drift rule would minimize $c_{\mathrm{aff}}(a)R_k + d_{\mathrm{aff}}(a)$. Such a rule can prefer a large q or b even when its resource cost leaves too few future contractions. Eq. (22) instead composes the drift over the action-dependent horizon in Eq. (7).

V. Learning-Aware Dependence Adaptation

A. Adaptive information lower bound

When θ_j is unknown, the controller may choose a predictable sequence $A_k = (q_k, b_k, \eta_k)$ and stop at a budget-bounded stopping time τ. Rotating X_k into its normalized agent mean $Y_k = q_k^{-1/2}\mathbf{1}^\top X_k$ and orthogonal contrasts shows that only Y_k contributes to the likelihood ratio. Under hypothesis j, a scalar Kalman filter gives the predictive mean $m^-_{j,k}$ and variance $P^-_{j,k}$, with innovation variance $V_{j,k} = 1 + q_k P^-_{j,k}$.

Let $Z_k = (m^-_{0,k}, P^-_{0,k}, m^-_{1,k}, P^-_{1,k})$. The conditional directional information is

$$\mathcal{I}_{01}(Z_k, A_k) = \frac{1}{2}\Bigg[\log\frac{V_{1,k}}{V_{0,k}} - 1 \\ + \frac{V_{0,k} + q_k(m^-_{0,k} - m^-_{1,k})^2}{V_{1,k}}\Bigg]. \tag{26}$$

The dependence of information on the resource action is already visible under a fixed probe. Let $T_n(r)$ be the $n \times n$ Toeplitz matrix with entries $[T_n(r)]_{uv} = r^{|u-v|}$, and let $\zeta_1, \ldots, \zeta_n$ be its eigenvalues.

Proposition 3 (Exact fixed-probe information)**.** *For a fixed probe (q, b, n) with $r = \lambda^b$, the directional Kullback–Leibler (KL) divergence is*

$$\mathrm{KL}(P_0^{q,b,n} \| P_1^{q,b,n}) = \frac{1}{2}\sum_{\ell=1}^{n}\left[x_\ell - 1 - \log x_\ell\right], \tag{27}$$

where $x_\ell = (1 + q\theta_0\zeta_\ell)/(1 + q\theta_1\zeta_\ell)$. The Bhattacharyya exponent used by the upper rule is

$$I_B(q, b, n) = \frac{1}{2}\sum_{\ell=1}^{n}\log\frac{1 + q(\theta_0 + \theta_1)\zeta_\ell/2}{\sqrt{(1 + q\theta_0\zeta_\ell)(1 + q\theta_1\zeta_\ell)}}. \tag{28}$$

They follow by diagonalizing the common-mode covariance $I + q\theta_j T_n(r)$; the orthogonal spatial contrasts have identical laws under both hypotheses. Participation scales common-mode eigenvalues rather than creating independent temporal records.

Theorem 3 (Stopped adaptive information identity)**.** *If the action kernel is predictable and common to both hypotheses and τ is bounded by Eq.* (6)*, then*

$$\mathrm{KL}(P_0^\pi|\mathcal{F}_\tau \,\|\, P_1^\pi|\mathcal{F}_\tau) = \mathbb{E}_0^\pi\sum_{k=1}^{\tau}\mathcal{I}_{01}(Z_k, A_k). \tag{29}$$

The reverse identity holds after exchanging the hypotheses.

Supplementary Appendix C proves cancellation of predictable action probabilities in the stopped likelihood. Consequently, any rule whose two directional decision errors are at most $\delta < 1/2$ must satisfy

$$\mathbb{E}_j^\pi\sum_{k\leq\tau}\mathcal{I}_{j,1-j}(Z_k, A_k) \geq \mathrm{kl}(1-\delta, \delta). \tag{30}$$

Together with Eq. (6), this gives an occupation-measure lower bound that prices participation, spacing, stopping, and delay without reducing the adaptive Markov experiment to independent action counts.

For a measurable belief-state set G, define the stopped occupation measure

$$\nu_j(G, a) = \mathbb{E}_j^\pi\sum_{k\leq\tau}\mathbf{1}\{Z_k \in G, A_k = a\}. \tag{31}$$

Let $\Delta_j(z,a)$ denote the certified one-step learning opportunity lost by using action a for identification instead of the known-instance learning action.

Proposition 4 (Adaptive opportunity lower bound)**.** *Every predictable, budget-feasible, δ-correct identification policy has expected opportunity cost at least*

$$\inf_{\nu\in\mathcal{M}_j}\int \Delta_j(z,a)\nu(dz,da), \tag{32}$$

where $\mathcal{M}_j$ contains the controlled Kalman-flow constraints, both resource constraints in Eq. (6)*, and the directional information constraint*

$$\int \mathcal{I}_{j,1-j}(z,a)\nu(dz,da)\geq \mathrm{kl}(1-\delta,\delta). \tag{33}$$

Supplementary Appendix D proves the belief flow: an early (q,b) changes later predictive variances, so independent action counts cannot replace controlled occupations.

B. Learning-aware upper rule

A fixed probe design $d=(q_d,b_d,n_d)$ consumes $C_d^{\mathrm{msg}}=n_d(h+q_d)$ and $C_d^{\mathrm{env}}=n_d b_d$. Its length is chosen so the exact Bhattacharyya exponent is at least $\log(1/\delta)$, which bounds each directional error by δ. Write $B_d=(B_{\mathrm{msg}}-C_d^{\mathrm{msg}},B_{\mathrm{env}}-C_d^{\mathrm{env}})$. We use $B_d^D=B_d$ to mark the delayed learner; Eq. (7) charges its declared (ℓ,u) once.

The probe is disjoint from learning samples and pays $n_d(h+q_d)$ messages and $n_d b_d$ clock units. Learning starts from the declared parameter history, and its moment and mixing certificates must hold uniformly conditional on the probe history. All selectable actions, including a wrong-instance commitment, require valid certificates under either true instance.

For each probe d, freeze selected actions $\widetilde{a}_j$ before observing its data and set $\overline{R}_j^{\mathrm{sel}}=\overline{R}_j(\widetilde{a}_j;B_d^D,D)$. Selections may use scalar candidates or a shielded central score; admission uses the upper certificates of the actual frozen selections. With $\overline{R}_j^{\mathrm{all}}$ the fallback certificate, define

$$\begin{aligned}
C_j^{\mathrm{probe}}(d)&=\overline{R}_j^{\mathrm{all}}(B_d,0)-\overline{R}_j^{\mathrm{all}}(B,0),\\
C_j^{D}(d)&=\overline{R}_j^{\mathrm{all}}(B_d^D,D)-\overline{R}_j^{\mathrm{all}}(B_d,0)\\
&\quad-\left[\overline{R}_j^{\mathrm{all}}(B,D)-\overline{R}_j^{\mathrm{all}}(B,0)\right],\\
G_j(d)&=\overline{R}_j^{\mathrm{all}}(B_d^D,D)-\overline{R}_j^{\mathrm{sel}},\\
W_j(d)&=\max_{v\neq j}\left[\overline{R}_j(\widetilde{a}_v;B_d^D,D)-\overline{R}_j^{\mathrm{sel}}\right]_+.
\end{aligned} \tag{34}$$

Thus $C_j^{\mathrm{probe}}+C_j^D=\overline{R}_j^{\mathrm{all}}(B_d^D,D)-\overline{R}_j^{\mathrm{all}}(B,D)$ exactly. The signed C_j^D compares delay penalties against the same fallback; the auxiliary zero-delay certificate suppresses (D,u,ℓ) together. The low-instance safety slack is

$$\epsilon_{\mathrm{safe}}(d)=\left[C_0^{\mathrm{probe}}(d)+C_0^D(d)-G_0(d)+\delta W_0(d)\right]_+. \tag{35}$$

The learning-aware controller probes only when

$$\begin{aligned}
C_1^{\mathrm{probe}}(d)+C_1^D(d)+\delta W_1(d)&<G_1(d),\\
\epsilon_{\mathrm{safe}}(d)&\leq\bar{\epsilon}.
\end{aligned} \tag{36}$$

Otherwise it executes the all-agent fallback without spending a probe. After an admitted probe, it commits to $\widetilde{a}_{\widehat{j}}$.

Any publicly specified certified fallback can replace the all-agent reference in Eq. (34).

Theorem 4 (Main finite-budget learning-value guarantee)**.** *Suppose Assumption 1 holds for both instances, the probe is feasible under Eq.* (6)*, and its directional errors are bounded by δ. Require each post-probe action's certificate to hold conditional on the probe history, from the declared learning initialization with a common bound R_0. Every design admitted by Eq.* (36) *satisfies*

$$\mathbb{E}_j\|e_{\mathrm{LA}}\|^2\leq\overline{R}_j^{\mathrm{sel}}+\delta W_j(d), \tag{37}$$

has a high-instance risk upper bound strictly below the full-budget all-agent certificate, and has low-instance risk at most $\overline{R}_0^{\mathrm{all}}(B,D)+\bar{\epsilon}$. If no design is feasible or admitted, the controller returns the all-agent action and incurs no probing cost.

Supplementary Appendix E proves the conditional risk decomposition.

C. Same-stream risk transfer

Disjoint probe and learning observations can remain dependent through the common state at the probe's terminal time T. For the Gaussian model in Eq. (3), a fixed probe gives $C_T\mid\mathcal{H}_d\sim\mathcal{N}(m_j,P_j)$, with deterministic P_j and $\mathbb{E}_j m_j^2=s_j^2:=\theta_j-P_j$. For $\bar{X}_{q,t}=q^{-1}\sum_{i=1}^q X_{i,t}$, freeze two downstream linear programs with deterministic coefficients before probing:

$$Z_a=d_a+\sum_{t=1}^{N_a}\omega_{a,t}\bar{X}_{q_a,T+b_a t},\quad k_a=\sum_{t=1}^{N_a}\omega_{a,t}\lambda^{b_a t}. \tag{38}$$

Write $J_{j,a}=\mathbb{E}_j Z_a^2$ for a deterministic commitment to a on the stationary stream. For the correct program a_c and wrong program a_w, define

$$\begin{aligned}
\Delta J_j&=J_{j,a_w}-J_{j,a_c},\\
\Gamma_j^2&=4(d_{a_w}k_{a_w}-d_{a_c}k_{a_c})^2 s_j^2\\
&\quad+2(k_{a_w}^2-k_{a_c}^2)^2 s_j^4.
\end{aligned} \tag{39}$$

Theorem 5 (Same-stream actual-risk enclosure)**.** *For a fixed feasible probe with directional decision error at most δ, followed by either frozen program in Eq.* (38) *on the continuing stream,*

$$\mathbb{E}_j Z_{\widehat{a}}^2 \geq \left[J_{j,a_c} - \delta[-\Delta J_j]_+ - \sqrt{\delta}\Gamma_j\right]_+,$$
$$\mathbb{E}_j Z_{\widehat{a}}^2 \leq J_{j,a_c} + \delta[\Delta J_j]_+ + \sqrt{\delta}\Gamma_j. \tag{40}$$

Allow distinct valid directional bounds δ_j in Eq. (40) and define

$$U_j(d) = J_{j,a_c(d)} + \delta_j[\Delta J_j(d)]_+ + \sqrt{\delta_j}\Gamma_j(d),$$
$$F_j = J_{j,\mathrm{all}}(B), \qquad U_1(d) < F_1, \quad U_0(d) \leq F_0 + \bar{\epsilon}. \tag{41}$$

All post-probe horizons in J use the remaining budgets; F_j spends no probe cost. Select a feasible passing design deterministically before observing data, or return the full-budget fallback. Theorem 5 then gives actual high-instance improvement and the low-instance allowance in Eq. (41), without a uniform conditional Gaussian certificate. The supplementary material supplies the proof, exact learner weights, an executable gate, and its saved-result reconstruction. For a fixed likelihood-ratio decision between centered Gaussian probes with ordered covariances, the odd conditional-risk term cancels. Writing $\Delta_k = k_{a_1}^2 - k_{a_0}^2$, the remaining selection term is nonnegative under both instances if $\Delta_k \geq 0$ and nonpositive if $\Delta_k \leq 0$. The supplementary corollary gives one-sided penalties of order $\delta\log(1/\delta)$ and δ, respectively, and a concrete failure of independent-mixture risk transport. This specialization retains the general enclosure above for other decision geometries.

To compare with every fixed catalogue program at the full budget, fix an explicit prior $\pi \in (0,1)$ and let B_j bound the executed gate's actual risk, including probe cost. The sufficient separation condition is

$$(1-\pi)B_0 + \pi B_1 < \min_{a \in \mathcal{A}_{\mathrm{fix}}} \{(1-\pi)J_0(a) + \pi J_1(a)\}. \tag{42}$$

Linearity extends the comparison to regime-independent randomized fixed programs. The supplement decomposes this classical decision comparison into adaptation opportunity, remaining-horizon cost, and decision-event risk for the implemented learners.

D. Observable calibration and unknown initialization

A covariance probe must be computable from the learner's observations. Let its raw output be $Z = \mathbf{1}\nu + W_j$, where ν is the unknown target-induced level, (C_T, W_j) is centered jointly Gaussian under regime j, and $\mathrm{Cov}_j(W_j) = \Sigma_j \succ 0$. Subtracting the unknown level ν would require unavailable target information. Instead choose the first-difference matrix D and use the observable contrast $Y = DZ = DW_j$. The classical error-contrast construction [43] gives covariance $K_j = D\Sigma_j D^\top$ and Gaussian affinity

$$\rho_D = \det(K_0)^{1/4}\det(K_1)^{1/4}/\det[(K_0+K_1)/2]^{1/2}.$$

Each directional error of its zero-threshold likelihood rule is at most ρ_D. The common state at probe exit has conditional mean $m_j(Y)$ and explained variance $s_j^2 = \mathrm{Var}_j(m_j(Y))$, computed from this projected law. The supplement derives a linear-time two-filter implementation and a finite-dimensional information-cost bound for removing the target. Every probe observation still consumes both budgets.

Let the initial error be an unknown fixed $e \in [-r, r]$, with public $r > 0$. In this subsection j may range over any fixed finite collection of covariance models. Fix a regime prior $\pi_j > 0$, mixture law g, and model posterior $p_j(y)$. Assume that (a) the probe and every continuing program are frozen before observing Y; (b) e does not affect the contrast law; and (c) the continuing learners have deterministic linear weights and the common-state representation in Eq. (38). Write the deterministic initialization contribution as $d_a(e) = \zeta_a e$ and let $k_{j,a}$ denote its common-state transfer coefficient under model j. For each frozen post-probe program let $J_{j,a}^r = \zeta_a^2 r^2 + \mathrm{Var}_j(\sum_t \omega_{a,t}\bar{X}_t)$, including its remaining horizon. Its even conditional score is

$$L_a(y) = \sum_j p_j(y)\{J_{j,a}^r + k_{j,a}^2[m_j(y)^2 - s_j^2]\}. \tag{43}$$

This score uses neither the target nor the sign of the initial error. Write $F_a(y) = \sum_j p_j(y) J_{j,a}^r$, let u minimize L_a and c minimize F_a, and set $S_j(y) = m_j(y)^2 - s_j^2$. Both selectors use a fixed even tie rule. For an even selector write $R_\pi(v) = \sup_{|e|\leq r} R_{\pi,e}(v)$, and let φ be the standard normal density.

Theorem 6 (Conditional value and its state-information ceiling)**.** *For a fixed paid probe and finite scalar continuing-program catalogue with the same deterministic learner weights under each regime,*

$$\inf_v \sup_{|e|\leq r} R_{\pi,e}(v) = \mathbb{E}_g \min_a L_a(Y), \tag{44}$$

where v ranges over randomized Y-measurable selectors in the invariant-statistic information class. The selector u attains the value, and

$$0 \leq R_\pi(c) - R_\pi(u) \leq K,$$
$$K = 4\varphi(1)\sum_j \pi_j \,\mathrm{range}_a(k_{j,a}^2)s_j^2. \tag{45}$$

For the running mean and a stable averaged affine-delay learner with affordable horizons proportional to budget

scale B and uniform persistence and impulse bounds, $K = O(B^{-2})$.

Posterior-loss minimization is classical [39], [40], [42]; here endpoint averaging and evenness resolve the unknown initialization. The supplementary proofs bound each regime's centered-square term separately, allowing different persistences, and give the vector covariance dual using classical minimax interchange [44]. An explicit two-program running-mean construction attains a strictly positive B^{-2} state-value limit, showing that this order is sharp.

For the best fixed risks $I_{\mathrm{full}}, I_{\mathrm{post}}$ before and after payment and $R_{\mathrm{rob}} = \mathbb{E}_g \min_a L_a$, the paid gain decomposes as

$$I_{\mathrm{full}} - R_{\mathrm{rob}} = V_{\mathrm{reg}} + V_{\mathrm{state}} - C - T,$$

Here $V_{\mathrm{reg}} \geq 0$ is nominal regime value, $V_{\mathrm{state}} \in [0, K]$ is extra state value, C is the possibly signed remaining-horizon cost, and $|T| \leq K/2$ corrects nominal posterior scores to actual risk. The supplement proves this identity and the resulting paid-value bounds.

E. When identification outperforms all fixed prefixes

Theorem 6 separates knowledge of the covariance regime from knowledge of the current noise state. The latter has a second-order value for averaged learners; the former can change the leading risk coefficient. For a running mean with $0 \leq \lambda \leq 1 - \gamma < 1$ and budgets $(\lfloor \beta_m B \rfloor, \lfloor \beta_e B \rfloor)$ for fixed $\beta_m, \beta_e > 0$, define

$$a_a = \min\{\beta_m/(h + q_a), \beta_e/b_a\},$$
$$K_j(a) = \frac{q_a^{-1} + \theta_j(1 + \lambda^{b_a})/(1 - \lambda^{b_a})}{a_a}.$$

Let a_j^* minimize K_j, $K_\pi = \min_a \sum_j \pi_j K_j(a)$, $\Delta = K_\pi - \sum_j \pi_j K_j(a_j^*)$, and $A = \sum_j \pi_j [K_j(a_{1-j}^*) - K_j(a_j^*)]$. The fixed comparison permits every affordable training prefix and stop, at the same learner and step.

Corollary 2 (First-order separation and a no-gain boundary)**.** *For a finite catalogue and bounded delay, a paid target-invariant probe with fixed directional error bound $0 < \delta < 1/2$, followed by commitment to $a_{\hat{j}}^*$, certifies superiority to every fixed prefix and fixed random mixture for sufficiently large B if $\Delta > \delta A$. The certified gain is at least $(\Delta - \delta A)/B - C(n+1)/B^2$, where n is probe length and C is finite. If the regimes share a leading minimizing action, no selector from the maximal-horizon post-probe catalogue plus stop following a length-$O(\log B)$ probe gains more than $O(\log B/B^2)$ over the best fixed prefix.*

The supplement proves the prefix expansion, gives explicit conservative remainder constants and a sufficient budget threshold, and supplies a rational witness on the public grid with a positive leading margin of approximately 0.07765 at directional bound 0.025. Finite-budget admission uses the actual projected probe, exact horizons and signed risk bounds, rather than the asymptotic margin. The Gaussian branch admits a probe only when its prior-weighted upper risk is below the best full-budget fixed-prefix risk; otherwise it executes that fixed prefix without probing. The comparison concerns a declared learner and resource-program class, rather than unrestricted estimators. Here a fixed prefix repeats one constant participation–spacing pair; an arbitrary time-varying nonadaptive schedule is a different comparator.

F. Finite separation for averaging and TD critics

Consider $Q = h = 8$, initialization radius one, budgets $(16B, 4B)$ and $q, b \in \{1, 2, 4, 8\}$. The covariance box is $\theta_0 \in [0.04, 0.06]$, $\theta_1 \in [1.8, 2.2]$, independently varying $\lambda_j \in [0.15, 0.25]$, and prior $\pi \in [0.45, 0.55]$. Use eighteen paid unit-spacing observations from all actors, the target-invariant likelihood rule frozen at $(0.05, 2, 0.2)$, and commitments $(8, 4)$ or $(2, 2)$. For each learner, let I_{open} be its best worst-initialization prior risk over every predetermined schedule, stopping prefix and independent random mixture, with actual covariance parameters available to that comparator. For the averaged scalar TD(0) critic, use $e_t = \beta e_{t-1} + \eta \bar{X}_t$ with $\eta = 0.1$, $\mu = 1 - \gamma = 0.5$ and $\beta = 1 - \eta\mu$.

Theorem 7 (Robust finite-budget separation)**.** *Throughout the stated box, at $B = 8192$ the paid rule satisfies*

$$R_{\mathrm{paid}} \leq (1 - \epsilon) I_{\mathrm{open}},$$

with $\epsilon = 0.0254$ for the running mean with eight withheld updates and $\epsilon = 0.0186$ for the averaged TD critic with zero update delay. All probe and scheduled-update costs are included.

The comparison allows arbitrary schedule length and order, including nonperiodic schedules. Supplementary Section XXV proves the result by direct Gaussian error bounds and a period-free resource envelope. An explicit TD filter-boundary correction transfers the envelope beyond running means.

G. Finite-budget phase theorem

Let budgets grow on a public ray $(B_{\mathrm{msg}}(s), B_{\mathrm{env}}(s)) = (\lfloor s\beta_m \rfloor, \lfloor s\beta_e \rfloor)$. Consider a compact class with $0 < \theta_- \leq \theta_j \leq \theta_+$, $|\theta_1 - \theta_0| \geq \Delta_{\min} > 0$, $|\lambda| \leq 1 - \gamma$, bounded delay, finite $\mathcal{Q}, \mathcal{B}$, and a nonempty compact certified step domain. Require uniformly: (a) a fixed probe has $I_B(q, b, n) \geq ni_- > 0$; (b) $|\overline{R}_j(a; s) - K_j(a)/s| \leq M/s^2$, and removing cost v changes each certificate by at most $L(v+1)/s^2$

for $s \geq s_0(v+1)$; (c) the selected actions satisfy $K_1(\mathrm{all}) - K_1(\widetilde{a}_1) \geq g_{\min} > 0$ and $K_0(\widetilde{a}_0) \leq K_0(\mathrm{all})$; and (d) $W_j(d;s) \leq w_+/s$, with $\delta_0 w_+ \leq g_{\min}/2$. The Gaussian running mean with independently initialized learning data satisfies the risk expansion and budget-shift conditions (Supplementary Appendix F). This shrinking-risk class is distinct from the fixed positive-step affine recursion, whose certificate has a nonzero noise floor. Here v is the probe charge in budget-ray units, including bounded delay and rounding costs. Set $B_{\mathrm{oracle}} = 4M/g_{\min}$ and $B_\epsilon = (M + L + \delta_0 w_+)/\bar{\epsilon}$ for $\bar{\epsilon} > 0$. Let B_N be the smallest scale satisfying the two directional information constraints, and B_S the infimum of scales above which an admitted design exists at every larger scale.

Theorem 8 (Necessary–sufficient threshold sandwich)**.** *For every fixed admissible safety target and $0 < \delta \leq \delta_0 < 1/2$, there are class constants $C_{\mathcal{K}}, C'_{\mathcal{K}} < \infty$ such that*

$$B_N \leq B_S \leq C_{\mathcal{K}} B_N + C'_{\mathcal{K}}(B_{\mathrm{oracle}} + B_\epsilon + 1). \quad (46)$$

Below B_N, δ-correct identification is impossible, while above B_S the learning-aware controller achieves the guarantees of Theorem 4.

Supplementary Appendix F derives the expected-information and scale bounds, separating identification feasibility from downstream learning value. For a stable averaged affine learner on the same stream, $\sqrt{\delta}\Gamma_j = O(\sqrt{\delta}/s^2)$; the supplementary derivation extends the shrinking-risk interface and scale argument to the corrected upper rule.

VI. Implementation and Stochastic-Delay Interpretation

A. End-to-end controller

Algorithm 1 fixes the catalogue and confidence level before data. Cached system specifications or disjoint calibration provide moment/mixing certificates; the likelihood probe selects between prespecified instances.

B. Predictable confidence interfaces

Predictable upper bounds on moments and mixing turn calibration estimates into risk certificates. For a known finite augmented Markov model, the supplementary construction computes these quantities from its kernel and stationary law, including a worst-state total-variation bound at the actual TD-pair gap. For an unknown model, the following confidence interface still requires simultaneous coverage; point estimates alone do not provide it. For a calibration block ending before action selection, let $\widehat{K}_q$, $\widehat{\Omega}_q$, and $\widehat{r}_q$ be estimators and

Input: budgets B, lag profile, terminal charges (ℓ, u), action sets, step interval, allowance $\bar{\epsilon}$
for each dependence instance j and pair (q,b) **do**
  form δ, K_q, Ω_q, and the certified set $\mathcal{E}_{q,b}$
  construct scalar candidates and their valid upper certificates
end for
for each feasible prespecified probe d **do**
  charge its message, environment, and delay costs
  freeze shielded selections $\widetilde{a}_j$ and upper scores for the remaining budget
  evaluate C_j^{probe}, C_j^D, G_j, W_j, and ϵ_{safe}
end for
if some probe satisfies Eq. (36) **then**
  choose the admitted probe with smallest certified post-probe risk
  observe the charged probe, run the likelihood test, and commit to $\widetilde{a}_{\widehat{j}}$
else
  execute the certified fallback without probing
end if
Output: value parameter and complete resource ledger

Fig. 1. Learning-aware Lyapunov resource control.

let u_K, u_Ω, u_r be simultaneous error radii. The controller uses

$$K_q^+ = \widehat{K}_q + u_K, \quad \Omega_q^+ = \widehat{\Omega}_q + u_\Omega,$$
$$r_q^+ = \min\{1, \widehat{r}_q + u_r\}. \quad (47)$$

For each valid calibration realization, the upper quantities preserve the drift inequalities conditional on that calibration history. An action is excluded if $r_q^+ = 1$, or if the resulting μ_δ or step domain is invalid.

The upper values define a feasibility shield and risk envelope; a predictable central score may rank shield-passing actions. Write $a_a = a_\delta(\eta)$, $h_a = h_\tau(\eta)$, $c_a = c_{\mathrm{aff}}(a)$, $d_a = d_{\mathrm{aff}}(a)$, and $n = n_a$.

Corollary 3 (Robust certificate and selection sensitivity)**.** *Suppose $K_q^+ \geq K_q$, $\Omega_q^+ \geq \Omega_q$, and $r_q^+ \geq r_q$ jointly, and form c_a^+, d_a^+ from the upper values using $\lambda_a^+ = \sqrt{h_a/a_a^+}$ and the zero-delay limit. Every action satisfying $\mu_{\delta^+} > 0$ and $c_a^+ < 1$ obeys $\mathbb{E}\|e_{k+1}\|^2 \leq c_a^+ R_k + d_a^+$ and the replacement-block recurrence $(R_{k+2D+1} - R_\star^+)_+ \leq c_a^+(R_k - R_\star^+)_+$, where $R_\star^+ = d_a^+/(1-c_a^+)$. Moreover, if $c_a, c_a^+ \leq \bar{c} < 1$ and $d_a, d_a^+ \leq \bar{d}$, then*

$$|\mathcal{U}_a^+ - \mathcal{U}_a| \leq \left(nR_0 + \frac{\bar{d}}{(1-\bar{c})^2}\right)\Delta c_a + \frac{\Delta d_a}{1-\bar{c}} =: \varepsilon_a. \quad (48)$$

Consequently, minimizing the upper score incurs at most $2\max_a \varepsilon_a$ *certified risk beyond the exact-certificate oracle.*

Here $\Delta c_a = |c_a^+ - c_a|$ and $\Delta d_a = |d_a^+ - d_a|$ admit the explicit moment- and mixing-slack bounds in the supplementary material. Valid upper certificates control risk, while their tightness controls selection loss continuously.

Spectral, regeneration, or externally supplied mixing certificates must hold simultaneously at the conditioning scope of the drift bound. An unconditional squared-error bound also requires control of loss on the calibration coverage-failure event.

The supplementary material gives the associated stochastic delay differential equation and its contraction–forcing interpretation. The algorithm uses the discrete theorem with finite steps, integer delays, and exact rounding.

VII. Simulations

A. Protocol and evaluation questions

The simulations distinguish three effects: resource matching with known dependence, the value of paying to identify dependence, and the incremental value of conditioning on the current noise state. The fixed-model studies compare the executed gates with explicit fixed-program catalogues; separate policy-evaluation and cooperative-learning studies assess the resource-control interface without asserting a general optimizer guarantee. The affine action/convergence studies use scalarized cost $h + q + b$ per update with $(\ell, u) = (0, 0)$; their D controls stale parameter indices. EXP-016B uses separate budgets, $(\ell, u) = (0, D)$, and pays every scheduled block, including the final D withheld updates. Theorem 5 covers its consecutive Gaussian probe/learning blocks and averaged first-in, first-out learner through their exact linear weights.

Studies pair common random numbers, charge all messages and clock units, retain prespecified cells, and use same-seed reproductions or disjoint confirmation seeds. The tasks comprise seven-state Markov policy evaluation and MAPPO [47] on PettingZoo particles [48] and multi-agent MuJoCo (MaMuJoCo) locomotion [49]; full specifications are supplementary.

Parameter risk is $\|w - w^\star\|^2$ normalized by its value before learning. Return-estimation risk is $|V_w(s_0) - V_{w^\star}(s_0)|^2$ under the specified initial state and is normalized in the same way. Convergence AUC uses charged resource fraction, including probe costs. The supplement specifies paired-bootstrap estimands.

B. Known-dependence resource matching

The participation study crosses six participation levels, six correlations, three delays, three budgets and thirteen steps over 32 paired seeds. At $q = 32$, median effective participation falls from 30.996 for independence to 1.111 at correlation 0.9. A separate twelve-scenario joint-action study uses 32 new paired seeds and 1,152 runs. Every selected action is stable and budget feasible; every one-sided 99 percent upper mean lies below its bound. The supplement reports certificate tightness and complete designs; this study isolates known-dependence resource control.

C. Learning-value adaptation

The dependence-adaptation study uses 96 finite-threshold scenario families, two model layers, eight policies, and 192 independent evaluation seeds. The primary estimand is the paired risk difference between probing at B_{id} and applying the learning-value gate in the separation phase.

Relative to information-only probing, the learning-aware controller reduces Layer-A risk by 35.24 percent and affine delayed temporal-difference risk by 9.79 percent. Each reduction divides the paired risk improvement by the information-only risk. The simultaneous one-sided lower bounds on the absolute paired improvements are 0.1612 and 0.0577, respectively. Directional practical improvement occurs in 77 of 96 scenario families. All 96 prespecified design-level safety checks satisfy $S_{\mathrm{mean}} \le \epsilon_{\mathrm{safe}}$, and every trajectory obeys both resource budgets.

Applying Eq. (40) to all 104 saved families and eight policies uses exact affinities 0.03523–0.04975 as directional-error bounds; the probes' nominal 0.025 controls average error. On the original practical separation subset, the mean population improvement over information-only probing is enclosed by $[0.1607, 0.1952]$ in Layer A and $[0.0562, 0.0773]$ in Layer B; the observed means are 0.1767 and 0.0726. These bounds include selection dependence without an independent learning reset. The entire Layer-B separation region instead has mean enclosure $[-0.0049, 0.0252]$; the supplementary full-family analysis retains individual adverse comparisons and the original subset definition.

Risk reductions are 32.94 percent with delay, 43.72 percent when messages bind, and 30.13 percent when transitions bind: probing is costly when it consumes the remaining learner's limiting resource. The supplementary material retains the full reduction plot.

D. Direct validation of the actual-risk gate

A separate validation executes the gate based on Theorem 5: 16 Gaussian families, two learners, two

regimes, four budget scales, and 128 new paired seeds. The corrected gate, no-carryover ablation, information-only probe, all-agent fallback, fixed $(q, b) = (2, 1)$, and evaluation oracle give 196,608 trajectories, all semantically replayed. Its calibration density uses a known-zero pilot; unknown-target observability requires the distinct contrast construction above. The supplement reports all 16 macro comparisons and policy costs.

The exact-affinity probe meets directional bound 0.025; 113/128 designs admit probing. Fig. 2 averages all families. All eight high-regime improvement intervals over fallback are positive, but low-regime running-mean scale 32 gives -0.00229 with descriptive paired-bootstrap 95 percent interval $[-0.00323, -0.00136]$. There are 23/256 adverse cell means; the low-instance allowance is 0.1, not zero harm.

The carryover ablation coincides on all 32,768 paired rows, and fixed $(2, 1)$ has lower pooled risk but higher message cost. An outcome-known model analysis of all 128 designs certifies Eq. (42) for 29 designs at the stated balanced prior $\pi = 1/2$, all at scales 512 or 2048; no design receives a minimax certificate. The signed refinement changes no admission; its population enclosures are analytical, not confidence limits for sample means.

E. Conditional state value and the fixed-program comparison

A separate nine-policy study tests conditional-state selection with 64 new paired master seeds, both initial-error signs and the same 128 learner–family–budget designs. Its known-zero pilot yields normalized fixed-minus-conditional mean -0.993855, with descriptive 95 percent interval $[-1.015340, -0.973059]$. The posterior-marginal-minus-conditional mean is -1.12×10^{-5} with interval $[-1.06 \times 10^{-4}, 5.67 \times 10^{-5}]$. Thus neither practical fixed-policy superiority nor incremental state value is confirmed on this grid: low-budget probe cost dominates, consistent with the distinction in Theorem 6. The supplement reports all nine policies, costs and initialization-sign comparisons.

F. Observable probing against every fixed prefix

A prospectively specified study removes the known-zero pilot assumption and strengthens the no-probe benchmark to every affordable training prefix and stop for every original participation–spacing pair, at the same frozen learner and step. It uses 64 fresh paired master seeds, both hidden initialization signs and all 128 designs, giving 294,912 trajectories with full semantic replay. The target-invariant probe and signed prior-risk gate admit 29 designs and execute the strongest fixed prefix on the remaining 99. The registered normalized fixed-minus-gate mean is 0.004443, with one-sided 99.5 percent bootstrap lower bound -0.003177. The uncertified-probe-minus-gate mean is 0.170210, with lower bound 0.148124. The fixed comparison misses its registered confidence and 0.01 mean conditions, while the second contrast supports economic admission within this model grid. The supplement retains all adverse designs, raw pooled risk, resource costs and the coincident carryover ablation. An outcome-known population calculation using classical quadratic-form mixtures [50] predicts normalized gain 0.00875, also below 0.01; the supplement derives the exact continuing-risk functional and numerical qualifications. The constructive parameter-box guarantee in Theorem 7 is a separate population statement, not a replacement for this all-design primary comparison.

G. Convergence and return-estimation curves

The convergence study tests the complete joint controller on the seven-state policy-evaluation problem over a budget of 128,000 resource units. The prespecified grid crosses persistence $\{0, 0.9, 0.98\}$, correlation $\{0, 0.9\}$, and delay $\{0, 8\}$, producing twelve cells and 41 charged-resource checkpoints per trajectory. The fixed policies use $q \in \{1, 4, 16, 32\}$; within every cell, each fixed q receives its own certified spacing and step. This comparison isolates the value of choosing participation while allowing every policy to adapt its spacing and step.

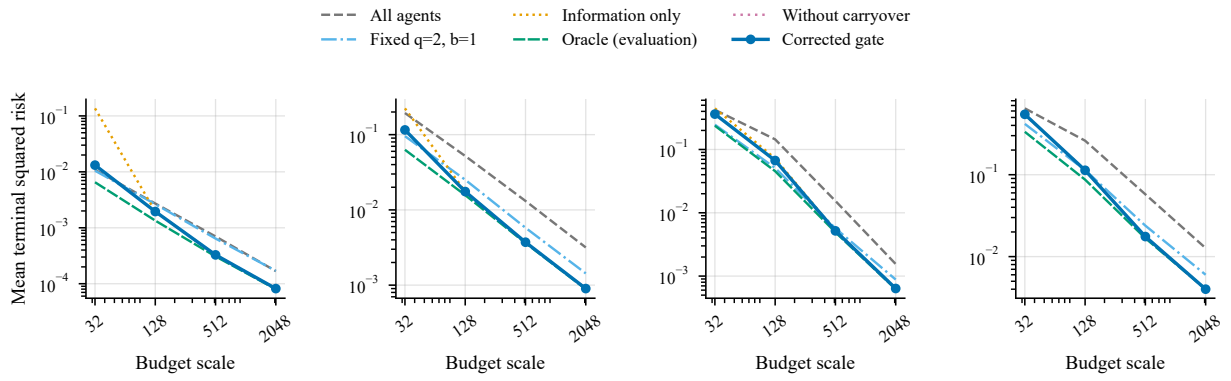
Sixteen development seeds select fixed $q = 4$; the locked comparison uses 64 disjoint confirmation seeds and 157,440 finite checkpoint rows.

Relative to fixed $q = 4$, the joint certificate has parameter-error AUC ratio 0.9009 with paired-bootstrap 95 percent interval $[0.8914, 0.9102]$. Its discounted-return-estimation AUC ratio is 0.8868 with interval $[0.8678, 0.9072]$. Terminal parameter and return-estimation errors decrease by 26.48 and 21.67 percent, respectively. Both AUC metrics improve in nine of twelve cells and tie in the remaining three; no cell is worse.

Supplementary Fig. 8 retains the complete persistence-0.9, zero-delay resource-indexed curves and confidence bands.

H. Transfer across cooperative policy-learning environments

Held-out, fully charged MAPPO studies select dependence-matched participation in all 32 controller runs. Against the strongest single fixed action, $q = 8$, normalized AUC improves by 2.319 percent in particles and the paired equal-regime mixture by 8.341 percent in locomotion. The supplement retains both complete protocols, fixed-participation comparisons, paired statistics, and unsmoothed return curves; these are not comparisons with general multi-agent optimizers.



(a) Running mean, low (b) Running mean, high (c) Delayed affine, low (d) Delayed affine, high

Fig. 2. Direct validation on new seeds: terminal squared risk averaged over all 16 families and 128 seeds per family, at each learner, regime, and budget scale. All six policies are shown; the corrected and no-carryover curves coincide exactly. Lines connect distinct budget experiments, not within-run learning checkpoints.

I. Ablations, complexity, and reproducibility

On 64 disjoint seeds, both confidence shields retain the nominal action in all twelve cells; all 24 stress-by-cell one-sided 99 percent upper means remain below the robust certificate. The supplement gives both stress levels and complete checkpoint counts.

The same study also isolates the finite-horizon objective. Against one-step myopic drift, terminal parameter risk falls by 12.20 percent; against a correlation-only participation rule, it falls by 64.22 percent. The respective ratios and paired-bootstrap 95 percent intervals are $0.878\ [0.836, 0.923]$ and $0.358\ [0.307, 0.412]$, both strictly below one.

Likelihood updates are scalar, and cached certificate search has the cost in Proposition 2, with spectral calibration charged separately.

The supplement retains the complete robustness and policy-learning reproducibility protocols.

VIII. Conclusion

Paid identification is useful only when its effect on the continuing learner compensates for the training horizon it consumes. For the specified Gaussian linear-program class, regime information can change the leading risk coefficient, whereas current-state information has a sharp second-order value for averaged learners. The fixed-prefix separation condition and finite-budget risk gate turn that distinction into a checkable comparison with constant nonadaptive programs. A constructive separation from every predetermined resource schedule holds for running means and averaged scalar TD critics throughout an explicit covariance and prior uncertainty box, with observations and scheduled updates fully charged. The Lyapunov resource score extends the design interface to affine delayed policy evaluation under certified moment and mixing bounds. The resulting theory and resource-matched simulations distinguish the value of identifying a dependence regime from the value of avoiding an uneconomic probe.

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